ALPHA-BETA PRUNING

```
def alpha_beta(depth, node_index, maximizing_player,
               values, max_depth, alpha, beta):

    if depth == max_depth:
        return values[node_index]

    if maximizing_player:
        best = float('-inf')

        for i in range(2):
            value = alpha_beta(
                    depth+1,
                    node_index * 2+i,
                    False,
                    values,
                    max_depth,
                    alpha,
                    beta
                )
            best = max(best, value)
            alpha = max(alpha, best)
            if beta <= alpha:
                break
        return best

    else:
        best = float('inf')
        for i in range(2):
            value = alpha_beta(
                    depth+1,
                    node_index * 2+i,
                    True,
                    values,
                    max_depth,
                    alpha,
                    beta
                )
```

```
        best = min(best, value)
        beta = min(beta, best)
        if beta <= alpha:
            break
    return best

values = [3,5,2,9,12,5,23,23]

max_depth = 3

result = alpha_beta(
    0,
    0,
    True,
    values,
    max_depth,
    float('-inf'),
    float('inf')
)
print("Optimal value:", result)
```

Output:

Optimal value: 12